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Buy the Weekend Dip?

Bitcoin Reversals After Extreme Sell-Offs

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Abstract: **Does Bitcoin rebound after an extreme one-hour sell-off, and is the rebound larger at weekends or when trading activity before the sell-off was low? This study tests both questions on hourly Binance BTCUSDT spot data. The primary rules and tests were fixed before outcomes were inspected; 2020 to 2023 was used for development, and January 2024 to July 2026 was held out for one primary historical evaluation. In both periods the average 12-hour log return after a negative shock was slightly negative (−0.13% and −0.10%), and none of H1 to H4 was supported held out. The all-shock rule returned −52.57% held out after assumed costs; the weekend rule, which gained 1.88% in development, returned −11.74% held out.**

1. Hypothesis Pitch

After an unusually large one-hour fall in BTCUSDT, I expect a partial recovery over the next 12 hours, with a larger rebound at weekends or after 24 hours of low trading activity. My central question is whether weekend timing adds information beyond the activity observed before the fall.

Temporary selling pressure could push prices down before easing, allowing a rebound. A fall driven by new information could instead continue. Lower preceding activity might accompany stronger temporary price pressure, but this is a proposed explanation: the hourly candles cannot identify its cause.

Previous evidence of short-term Bitcoin mean reversion and reversal after negative hourly shocks makes a rebound plausible [1, 2]. Evidence of stronger daily reversal strategy returns across cryptocurrency pairs when volume is low relative to its recent trend motivates testing activity as well [3]. These findings leave open whether pre-shock activity matters within BTCUSDT and whether weekend timing adds information beyond it.

In De Nicola's Bitstamp sample, weekend volume was about 20% below the weekly average [1], but that does not establish a weekend rebound. A quiet weekday could resemble a quiet weekend. I therefore compare weekend timing and activity separately, then ask whether the weekend association remains after adding activity and other controls. I measure activity using the USDT value traded before the shock. This records completed trading, but does not reveal the available orders, spreads or depth.

Four predictions were fixed before outcomes were inspected. I would set aside each unless the held-out data show:

1) a positive mean 12-hour return after negative shocks (H1);
2) a higher mean after weekend than weekday shocks (H2);
3) a higher return after lower pre-shock activity (H3);
4) a positive weekend association after controlling for activity, shock size, recent volatility and time of day (H4).

Each needs the expected sign and a Holm-adjusted one-sided p-value below 0.05. An unsupported test does not establish that the effect is absent; all planned analyses continue. Secondary H5 compares positive and negative shocks. Separately, I would reject the trading case if development gains failed held out or disappeared after costs.

2. Data Pipeline

Archives are downloaded and verified, cleaned and divided into contiguous segments. Past-only features identify shock candidates; chronological cooldown sets event acceptance. Counts and timestamp-only availability are audited before outcomes. Accepted negative events feed the event study and simulated trades. Development checks precede the freeze of rules and inputs and the held-out evaluation.

A. Data Source and Snapshot

The code downloads 82 official Binance BTCUSDT spot monthly archives [4], verifies each ZIP against Binance's published SHA-256 checksum and preserves it unchanged in `data/raw/`. Their hourly OHLC prices, volume and trade counts cover October 2019 to July 2026. The 6-, 12- and 24-hour analyses all use these one-hour candles. No credentials are needed.

B. Sample Periods

October to December 2019 warms up rolling features. Development covers 2020 to 2023; January 2024 to July 2026 is held out. Weekend and split labels use the unrounded UTC source close time.

C. Cleaning and Missing Data

Binance switched spot archive timestamps from milliseconds to microseconds on 1 January 2025 [4]. The parser enforces the expected unit, 12-column schema and numeric/OHLC consistency. Invalid values or duplicate/disordered timestamps stop validation; the snapshot contains none (Table I).

8 non-standard close-time rows coinciding with exchange interruptions are excluded. The cleaned grid has 43 missing hours, including these exclusions: 4 in warm-up, 39 in development and none held out, forming 19 contiguous segments. Manual raw-file checks retained all 39 distinct rows identified by the 20 largest returns and 20 largest close-to-next-open gaps.

No candle is imputed. Gaps invalidate crossing returns and outcomes and restart rolling histories. Bitcoin needs no split or dividend adjustment.

TABLE I
Data quality (left) and hours each calculation can use (right)

Raw archive, Oct 2019 to Jul 2026	Rows
Monthly files	82
Rows parsed	59,869
Excluded (non-standard close)	8
Duplicate / invalid rows	0 / 0
Clean rows	59,861
Missing hours	43

	Development	Held out
Observed clean hours	35,025	22,632
Hours with a valid shock score	25,979	22,632
Hours with every feature	15,323	22,632

D. Shock Events and Features

Features use information available at the shock close. The hourly log return is $r_t = \ln(C_t/C_{t-1})$, requiring a contiguous previous hour. The score $z_t = (r_t - \bar{r})/s$ uses the mean $\bar{r}$ and sample standard deviation s of the previous 720 returns (30 days), excluding r_t. A candle with $z_t < -2.5$ is a negative-shock candidate.

Candidates are processed chronologically: accept the first, then suppress same-sign candidates less than 24 hours after the last accepted event. This limits repeated signals from one sell-off. Future prices and outcomes never determine acceptance.

Pre-shock activity A_t is the USDT quote volume of the previous 24 hours, excluding the shock hour. Its rank P_t is the share of the previous 2,160 values of A (90 days) less than or equal to it. Each reference value needs 24 prior hours, so the full rank requires 2,184 uninterrupted hours. Regressions use $1 - P_t$; low activity means $P_t < 1/3$. Quote volume measures activity, not liquidity.

Controls are shock size $(-z_t)$, realised volatility of the previous 24 returns, and four six-hour UTC time-of-day blocks.

The outcome $\ln(C_{t+12}/C_t)$ is the 12-hour log return after the shock, available only when all 12 following hours are contiguous and within one split. Table II reconciles counts audited before outcomes.

TABLE II
SHOCK EVENTS AT EACH STEP

	Development	Held out
Negative candidates ($z < -2.5$)	387	384
Accepted after the 24-hour cooldown	235	222
weekend	35	27
weekday	200	195
low activity	25	64
activity rank unavailable	77	0
weekend and low activity	6	15
Complete 12-hour outcome (H1, H2)	233	221
Complete outcome and features (H3, H4)	158	221
Accepted positive events ($z > 2.5$)	236	216

Event acceptance and later eligibility are separate. Of the 235 accepted development negative shocks, 233 have complete 12-hour outcomes and 158 also have every controlled-regression covariate (Table II). Missing features do not change accepted event identities. The all-shock and weekend strategies require valid execution bars; low-activity and combined strategies additionally require an activity rank.

3. Strategy and Backtest

A. Key Results

TABLE III
KEY RESULTS AT THE PRIMARY COST (10 BP FEE AND 2 BP SLIPPAGE PER SIDE)

	Development, in-sample				Held out, out-of-sample			
Strategy	Trades	Net return	Sharpe	Max DD	Trades	Net return	Sharpe	Max DD
All shock	233	−57.47%	−0.957	−65.31%	221	−52.57%	−1.328	−57.10%
Weekend	34	1.88%	0.100	−14.34%	27	−11.74%	−0.435	−19.74%
Low activity	25	−6.37%	−0.302	−12.61%	64	−11.91%	−0.497	−19.14%
Combined	6	−10.10%	−0.697	−10.52%	15	−8.02%	−0.586	−11.86%
BTC buy and hold	1	486.25%	n/r	n/r	1	48.37%	n/r	n/r

Net return is the cumulative return after costs over the whole period. The Sharpe ratio uses daily returns, annualised with 365 days and a zero risk-free rate. The recorded daily series samples equity at midnight after any new entry, so a midnight entry cost is assigned to the preceding day. This boundary convention can slightly change Sharpe relative to a mark immediately before the new entry. Maximum drawdown (Max DD) is measured on the hourly equity curve. The held-out combined rule has 15 trades; its annualised Sharpe is descriptive and highly unstable. Buy and hold has a different exposure and is not risk-matched; n/r means not reported.

Table III gives the required metrics. At the primary settings, all four strategies lost money held out after costs. The weekend rule, the main candidate, returned 1.88% in development (Sharpe 0.100) and −11.74% out-of-sample (Sharpe −0.435, maximum drawdown −19.74%). Buying every shock returned −57.47% and −52.57%. Over the held-out period, simply holding Bitcoin returned 48.37%.

B. Trading Rules

When a qualifying negative-shock candle closes, the rule buys BTC at the next hour's open and sells at the close 12 hours after the shock (entry O_{t+1}, exit C_{t+12}). Each trade invests 100% of the account, without leverage, short selling or a stop-loss. One position is open at a time; otherwise the account holds USDT. Four separate accounts start at 1.0 and trade all shocks, weekend shocks, low-activity shocks, or shocks that are both. All four are reported. Missing entry-to-exit hours cancel the trade. Event-study log returns begin at the shock close; portfolio wealth compounds arithmetic net trade returns from next-open execution after costs.

C. Transaction Costs

Costs are charged on every fill. With fee f and slippage s per side, the buy price is $O(1+s)$, the sell price is $C(1-s)$, and each trade multiplies wealth by $(1-f)^2\, C(1-s)/\big(O(1+s)\big)$. The primary case assumes $f = 10$ bp and $s = 2$ bp, about 24 bp per round trip. A low case (5 bp fee, 1 bp slippage) and a high case (10 bp, 10 bp) were fixed in advance.

D. Statistical Tests

Each hypothesis is tested with an ordinary least-squares regression on shock events, separately by period. Model 0 estimates the mean outcome (H1). Model 1 adds a weekend indicator (H2). Model 2 adds the activity score, shock size, volatility and time-of-day controls, giving H3 (activity) and H4 (weekend after controls). Secondary Model 3 adds a weekend × activity interaction. HC3 standard errors account for unequal residual variance, but cannot rule out dependence between shocks. Holm's correction limits false positives across the four one-sided H1 to H4 tests. As a check on within-month dependence, a bootstrap resamples whole calendar months 10,000 times and refits each model.

E. Returns After a Shock

The rebound the hypothesis predicts does not appear on average. The mean 12-hour log return after a negative shock is −0.13% in development and −0.10% out-of-sample, against 0.061% and 0.020% for eligible non-candidate hours (Table IV). Medians are slightly positive (0.11% and 0.04%), so the negative means are pulled down by a tail of continued falls. The weekend group is positive in development (0.29%, $N = 34$) but negative out-of-sample (−0.22%, $N = 27$). The average path (Fig. 1) is small and noisy; in both periods it is below zero at 12 hours and stays positive only from hour 19 in development and hour 18 out-of-sample.

TABLE IV
12-HOUR LOG RETURN AFTER SHOCK EVENTS

	Development				Held out			
Group	N	Mean	Median	Hit rate	N	Mean	Median	Hit rate
All negative events	233	−0.13%	0.11%	53.6%	221	−0.10%	0.04%	50.2%
weekend	34	0.29%	0.30%	55.9%	27	−0.22%	−0.32%	40.7%
weekday	199	−0.20%	0.10%	53.3%	194	−0.08%	0.17%	51.5%
low activity	25	−0.02%	0.19%	52.0%	64	0.04%	−0.24%	46.9%
other activity	133	−0.18%	0.10%	54.1%	157	−0.15%	0.21%	51.6%
weekend and low activity	6	−1.53%	−1.27%	33.3%	15	−0.32%	−0.52%	40.0%
Positive events	235	0.26%	0.05%	50.6%	216	0.20%	0.15%	53.7%
Non-candidate hours (context only)	25,463	0.061%	0.027%		22,237	0.020%	0.045%	

Hit rate is the share of events with a positive 12-hour return. Non-candidate hours satisfy $z_t \geq -2.5$, including positive extremes; their windows overlap, so this row is context, not a formal comparison.

The secondary event-study means are also uncertain. At 6 hours they are −0.06% ($N = 235$, HC3 95% CI [−0.31%, 0.19%]) in development and 0.00% ($N = 222$, CI [−0.21%, 0.22%]) held out; at 24 hours they are 0.21% ($N = 233$, CI [−0.24%, 0.65%]) and 0.13% ($N = 221$, CI [−0.25%, 0.50%]).

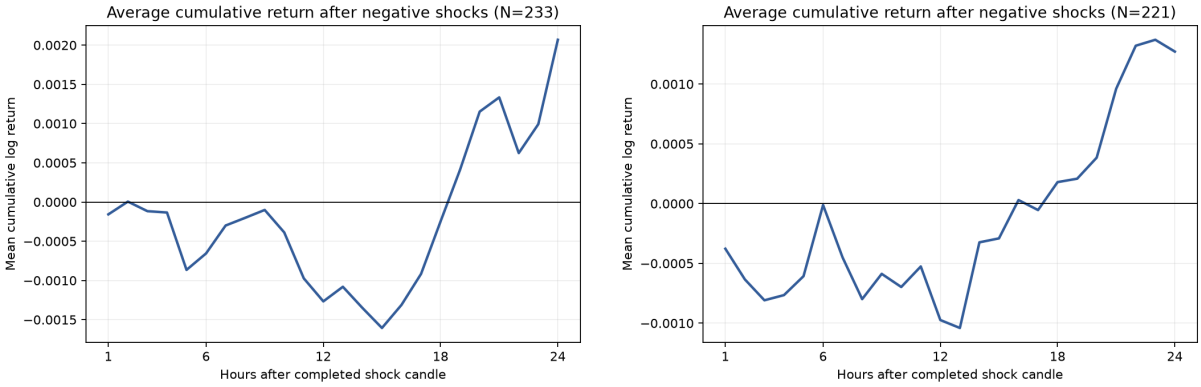
The direct group comparisons in Table V use indicator regressions with HC3 uncertainty and a calendar-month block bootstrap. The group medians and counts are in Table IV; the combined group is sparse, especially in development ($N = 6$). Its development block interval excludes zero while the HC3 interval includes it, so that small-cell comparison is not a stable finding.

F. Hypothesis Test Results

None of H1 to H4 is supported in the held-out period (Table VI). All four primary HC3 intervals include zero in both periods, as do their month-block bootstrap intervals. The two weekend estimates change sign between the periods. The weekend difference (H2) is 0.49% in development but −0.14% out-of-sample. After controls (H4), the development weekend coefficient falls slightly, from 0.42% to 0.28% on the same

TABLE V
12-HOUR GROUP DIFFERENCES IN MEAN LOG RETURN

Period	Difference	N_1	N_0	Estimate	HC3 95% CI	Block 95% CI
Development	Weekend minus weekday	34	199	0.49%	[−0.42%, 1.41%]	[−0.28%, 1.42%]
Development	Low minus other activity	25	133	0.16%	[−0.79%, 1.10%]	[−0.75%, 0.98%]
Development	Combined minus remaining	6	152	−1.43%	[−3.79%, 0.93%]	[−4.11%, −0.04%]
Held out	Weekend minus weekday	27	194	−0.14%	[−1.27%, 0.99%]	[−1.26%, 0.93%]
Held out	Low minus other activity	64	157	0.20%	[−0.37%, 0.76%]	[−0.34%, 0.75%]
Held out	Combined minus remaining	15	206	−0.24%	[−1.38%, 0.91%]	[−1.63%, 0.91%]



(a) Development, $N = 233$

(b) Held out, $N = 221$

Fig. 1. Average cumulative log return in the 24 hours after a negative shock.

events; out-of-sample it moves from −0.14% to −0.40%. The activity coefficient (H3) is 0.01% and 0.42%, with intervals of [−1.89%, 1.91%] and [−1.00%, 1.85%]: too wide to show any reliable association.

TABLE VI
MAIN HYPOTHESIS TESTS

		Development				Held out				
	Test	N	Estimate	95% CI	p	N	Estimate	95% CI	p	Supported
H1	Mean return	233	−0.13%	[−0.47%, 0.21%]	1.000	221	−0.10%	[−0.38%, 0.19%]	1.000	No
H2	Weekend	233	0.49%	[−0.42%, 1.41%]	0.578	221	−0.14%	[−1.27%, 0.99%]	1.000	No
H3	Activity score	158	0.01%	[−1.89%, 1.91%]	1.000	221	0.42%	[−1.00%, 1.85%]	1.000	No
H4	Weekend, controlled	158	0.28%	[−1.00%, 1.55%]	1.000	221	−0.40%	[−1.60%, 0.79%]	1.000	No

H1 is the mean 12-hour log return (Model 0); H2 the weekend minus weekday difference (Model 1); H3 the activity score and H4 the weekend coefficient, both in Model 2. Intervals are two-sided 95% HC3 intervals; p is the Holm-adjusted one-sided value. Only the held-out columns decide support.

Table VII gives HC3 standard errors, raw one-sided p-values, block intervals and resample counts. HC3/Holm is primary; the bootstrap checks robustness.

The secondary results lie outside the Holm family, use two-sided unadjusted tests, and are descriptive only. The Model 3 interaction is negative in both periods: −4.53% in development (HC3 95% CI [−8.51%, −0.55%], block CI [−8.47%, −1.40%], $p = 0.026$), based on only 6 weekend low-activity events, and −2.69% out-of-sample (HC3 CI [−5.37%, −0.02%], block CI [−5.94%, 1.35%], $p = 0.049$). The held-out HC3 interval barely excludes zero, while its block interval includes zero; this pattern is uncertain and requires a new pre-specified test.

H5 compares 12-hour returns after positive and negative shocks. The positive-minus-negative coefficient, conditional on absolute standardised shock size and the other specified covariates, is 0.44% in development (HC3 CI [−0.20%, 1.07%], block CI [0.05%, 0.85%], $p = 0.178$) and 0.29% held out (HC3 CI [−0.13%, 0.70%], block CI [−0.04%, 0.65%], $p = 0.173$). The development HC3 and block intervals

TABLE VII
ADDITIONAL INFERENCE DETAILS FOR H1 TO H4

Period	Test	HC3 SE	Raw p	Holm p	Block 95% CI	Draws A/V/E/R
Development	H1	0.17%	0.768	1.000	[−0.43%, 0.17%]	10000/10000/0/0
Development	H2	0.46%	0.144	0.578	[−0.28%, 1.39%]	10000/10000/0/0
Development	H3	0.96%	0.495	1.000	[−1.00%, 1.18%]	10000/10000/0/0
Development	H4	0.65%	0.335	1.000	[−1.02%, 1.69%]	10000/10000/0/0
Held out	H1	0.15%	0.749	1.000	[−0.29%, 0.11%]	10000/10000/0/0
Held out	H2	0.57%	0.598	1.000	[−1.25%, 0.91%]	10000/10000/0/0
Held out	H3	0.73%	0.279	1.000	[−0.71%, 1.61%]	10000/10000/0/0
Held out	H4	0.61%	0.746	1.000	[−1.49%, 0.68%]	10000/10000/0/0

Draws A/V/E/R are attempted/valid/empty/rank-deficient month-block bootstrap replicates. The target is 10,000 valid draws per estimate, with at most 20,000 attempts. Standard errors and intervals use return-per-unit coefficients displayed as percentages.

disagree about zero; held-out evidence does not establish asymmetry. With one cooldown shared by both signs, the coefficient is positive in both periods (0.87%, $p = 0.015$ in development; 0.26%, $p = 0.251$ held out), but statistically distinguishable from zero only in development.

G. Backtest and Cost Sensitivity

Most rules lose before costs, and costs reduce every strategy's return (Table VIII, Fig. 2).

TABLE VIII
CUMULATIVE RETURN BEFORE COSTS AND UNDER EACH COST CASE

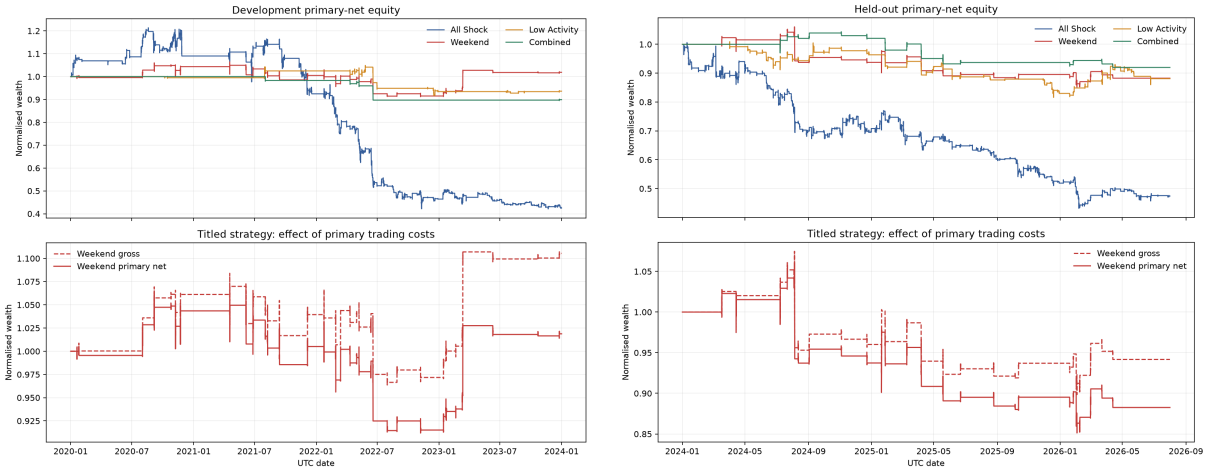
Period	Strategy	Gross	Low cost	Primary	High cost	Break-even
Development	All shock	−25.58%	−43.74%	−57.47%	−70.70%	none
	Weekend	10.54%	6.12%	1.88%	−3.52%	14.7 bp
	Low activity	−0.58%	−3.52%	−6.37%	−10.04%	none
	Combined	−8.79%	−9.45%	−10.10%	−10.96%	none
Held out	All shock	−19.37%	−38.15%	−52.57%	−66.70%	none
	Weekend	−5.83%	−8.83%	−11.74%	−15.47%	none
	Low activity	2.72%	−4.88%	−11.91%	−20.49%	2.1 bp
	Combined	−4.65%	−6.35%	−8.02%	−10.20%	none

Break-even is the cost per side at which the strategy ends where it started; "none" means it loses even at zero cost.

Held-out all-shock return falls from −19.37% gross to −52.57% net. The development weekend rule breaks even at 14.7 bp per side, while the only held-out rule positive gross, low activity (2.72%), breaks even at 2.1 bp.

H. Robustness Checks

Each robustness check was fixed in advance and changes one setting at a time (Table IX). In development, 4 of the 7 threshold, horizon, delay and cooldown variants of the weekend rule are positive, but the sign flips with a one-hour entry delay or a different cooldown, and removing a single event moves the result anywhere from −5.51% to 8.98%. Out-of-sample, 2 of the 7 variants are positive: the −3.0 threshold, with only 16 trades, and the 48-hour cooldown. The rule loses money in each of the 3 held-out calendar years, and removing any single event still leaves it between −15.17% and −2.89%. For H1, the confidence interval includes zero under every threshold, horizon and cooldown in both periods.



(a) Development (b) Held out

Fig. 2. Net equity of the four rules at the primary cost (top panels), and the weekend rule before and after costs (bottom panels).

TABLE IX
WEEKEND RULE WHEN ONE SETTING IS CHANGED (NET RETURN AT THE PRIMARY COST)

	Development		Held out	
Setting	Net return	Trades	Net return	Trades
Primary rule	1.88%	34	−11.74%	27
Shock threshold −2.0	7.97%	60	−7.85%	47
Shock threshold −3.0	6.91%	20	1.38%	16
Hold 6 hours	2.99%	35	−6.80%	27
Hold 24 hours	16.25%	34	−5.40%	27
Enter one hour later	−7.13%	34	−8.58%	27
Cooldown 12 hours	−12.34%	41	−9.23%	30
Cooldown 48 hours	−2.51%	29	1.99%	24
Signals inside the weekend	−2.55%	20	0.63%	12
Signals whose hold reaches Monday	4.54%	14	−12.30%	15
Leave one event out (range)	−5.51% to 8.98%		−15.17% to −2.89%	
Leave one year out (range)	−8.49% to 11.89%		−10.48% to −5.84%	

I. Interpretation

I take three lessons from the results. First, there is no reliable positive mean return at the fixed 12-hour horizon in this Binance sample. Earlier reversal studies used different venues, periods and event definitions, so this does not directly refute them. Second, the development weekend gain was fragile: it rested on 34 trades, its five largest trades made up 42.6% of total absolute profit and loss, and its sign changed with the cooldown or entry time. The held-out loss confirms that this gain did not persist. Third, at about 24 bp per round trip, the assumed costs leave little room for a small rebound to become a net trading gain.

4. METHODOLOGY DEFENSE

A. Universe

I chose BTCUSDT because it trades through weekends and Binance provides a public, checksummed history. Hourly candles define a signal from a completed bar and entry at the next open; they do not establish actual fills. This tests one surviving pair on one exchange, so findings may differ for other coins or venues.

B. Train/Test Split

I used 2020 to 2023 to develop the rule and kept the next 31 months, with 221 usable shocks, for evaluation. A random split would mix nearby shocks and let later market conditions influence choices made for earlier ones. The chronological split asks whether the same rule survives a later period. If the market changed after 2023, that limits how far the development result carries forward.

C. Signal Construction

A fixed price-drop cut would flag many hours in volatile years and few in calm years. I instead scale each return by the previous 30 days, so the -2.5 threshold describes an unusual move relative to recent conditions; it yields 235 development and 222 held-out events. I fixed 2.5 before inspecting outcomes as a convention for relative extremeness, with 2.0 and 3.0 as sensitivity checks. The cutoff is not claimed to be optimal. The 24-hour cooldown keeps the first candidate from a sell-off. Counting every extreme hour would treat one episode as several events, while choosing the largest would require later prices. I chose 12 hours as a short window in which temporary pressure might fade, with 6 and 24 hours fixed as checks.

D. Transaction Cost Assumptions

Next-open fills follow the completed shock and remain a simulation assumption. The 10 bp fee is fixed; 2 bp slippage allows for spread and delay without order-book estimates. Higher costs worsen every return, and a one-hour delay reverses the development weekend gain (Table IX).

E. Activity Measure

I use prior 24-hour USDT quote volume to summarise a full day before the shock. Ranking against 90 days provides a longer recent benchmark and adjusts for changing volume levels. H3 uses the continuous score $1 - P_t$; the pre-set $P_t < 1/3$ cutoff defines descriptive groups and trading filters. Trade count is an alternative check. Neither proxy measures order-book depth; high volume can accompany poor liquidity.

F. Statistical Inference

I refit the weekend-only model on Model 2's complete-case events, fixing sample composition when adding activity and controls. HC3 allows unequal residual variance; Holm guards H1 to H4 against chance findings. Month-block resampling probes within-month dependence; longer dependence may still leave intervals too narrow. Unmeasured news can affect shocks and subsequent returns, so coefficients describe associations, not causes.

G. Limitations

Sparse groups (6 and 15 weekend low-activity events), a few influential shocks, missing order-book data, and unmodelled Binance and USDT risks limit inference and execution claims.

5. Pitfall Checklist

A. Look-ahead bias

I checked features, events and trades separately. Changing later observations leaves earlier shock, activity and volatility inputs unchanged. Rolling histories exclude the shock return and volume; future outcomes cannot change cooldown decisions. Tests verify next-open entry after a completed shock, exact exits, gap rejection, unrounded calendar labels and split-contained outcomes. No future prices or returns enter signal construction. Trades crossing missing-data gaps are excluded retrospectively for evaluability, which limits the backtest's realism.

B. Survivorship bias

There is no retrospectively filtered asset universe: the study preselects one pair. Bitcoin and Binance nevertheless survived while other coins and venues failed. A collapsing coin might have more permanent sell-offs, so the result cannot be generalised to all crypto assets.

C. Overfitting/data-snooping

One primary setting, four trading rules and 12 robustness and diagnostic families were fixed before outcomes. No grid search or post-result tuning was used; corrections are logged. Holm adjusts H1 to H4 for

multiple testing. Every frozen variant was run. The 1.88% development weekend gain failed held out.

D. Transaction costs and slippage

The assumed costs are 10 bp fees and 2 bp slippage per side. Low and high cases and delayed entry test sensitivity; market impact is unmodelled. Held-out all-shock return falls from −19.37% gross to −52.57% net. In that period, only the low-activity rule is positive before costs, breaking even at 2.1 bp per side; all four primary rules lose net.

E. Out-of-sample validation

Development used 2020 to 2023; January 2024 to July 2026 was held out. Before access, an outcome-free audit flagged sparse cells and code, configuration, inputs and development results were frozen. One primary evaluation was accepted. Weekend return fell from 1.88% (Sharpe 0.100) to −11.74% (Sharpe −0.435). This is historical pseudo-out-of-sample evidence, not live performance.

6. Efficiency Touchpoint

The notebook benchmark identifies month-block regression bootstrapping as the bottleneck: ten development estimates required 100,000 refits and about 20 seconds ($N \leq 300$, $k \leq 9$), versus 4 seconds for features. Each draw samples whole months, rebuilds inputs and refits. For S estimates, B draws, N events and k regressors, time is about $O(SBNk^2)$; working memory per bootstrap is $O(Nk + B)$ for inputs and coefficient draws. Building each model matrix once and resampling its rows could reduce repeated work. The current run is practical and reproducible.

7. Risk Touchpoint

The account takes one unleveraged BTC position at a time, using 100% of capital and holding USDT between trades. It remains concentrated in BTC, Binance and USDT. Held-out weekend exposure is 1.4% of the time, yet maximum drawdown is −19.74%; brief exposure does not ensure small losses. Five trades account for 49.2% of absolute profit and loss. Lasting negative information could sustain a decline throughout the 12-hour hold while spreads widen. With no stop-loss, the account bears that decline. Full-capital sizing is a research convention.

8. Reflection

Cleaning was harder than the statistical tests: interruptions, a timestamp-unit switch and irregular closes needed explicit handling. Requiring complete covariates reduced the development sample from 233 outcome-eligible shocks to 158 controlled-model events. The reproducible pipeline, validation checks and frozen rules made the findings auditable.

I expected a positive 12-hour mean, but the development and held-out means were slightly negative. The low-activity weekend idea was unsupported; its exploratory interaction pointed the other way, with uncertain held-out robustness. I learned how convincing a 34-trade gain could look when a few trades drove it, and underestimated the effect of a 24 bp round trip. A plausible explanation and development profit must survive uncertainty, later data and costs before I can trust a rule.

Next I would pre-register the unchanged rules on other pairs and exchanges, and test whether order-book depth explains activity associations. This rule showed no held-out net edge under the assumed costs.

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